

AI Fluency: Framework & Foundations — Workflow Audit Report

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1. 15 Real-Week Recurring Tasks Audit Table (Ethan Mollick Framework)

#	Recurring Task Description	Context	Time	Quadrant	One-Line Rationale
01	API Auth & RBAC Security Sign-Off	Work	2.5h	Just Me	Final architectural security decisions and threat modeling require human accountability and ethical sign-off.
02	Junior Developer Mentoring	Work	3.0h	Just Me	Mentorship and constructive interpersonal communication require emotional intelligence and human empathy.
03	System Architecture & Schema Planning	Side/Work	2.0h	Just Me	Conceptual synthesis and strategic goal-setting must originate from human intuition and project vision.
04	Express.js Route & Controller Creation	Work	4.0h	Delegate (Review)	AI generates syntax-accurate handlers rapidly, but human review is mandatory for error boundaries & business rules.
05	SQL/Prisma Query Optimization	Work	2.0h	Delegate (Review)	AI excels at syntactic SQL restructuring, but query plans must be benchmarked against live data volumes.
06	OpenAPI/Swagger Doc Generation	Work	1.5h	Delegate (Review)	AI accurately maps route signatures to OpenAPI specs, but requires review for edge-case auth header descriptions.
07	Dockerfile Multi-Stage Build Optimization	DevOps	1.5h	Delegate (Review)	AI generates standard layers efficiently, but security stripping and Alpine dependency compat require manual check.
08	Debugging Async Race Conditions & Pooling	Work	3.5h	Collaborate	Pairing with AI as an analytical sounding board accelerates hypotheses while human intuition guides experiments.
09	Designing RAG Embedding & Chunking Strategy	Side (RAG)	3.0h	Collaborate	Iterating on embedding architectures benefits from dialogue where AI suggests algorithms and human tests quality.
10	Refactoring Legacy Controllers to Services	Work	2.5h	Collaborate	Refactoring works best as an interactive dialogue where AI drafts structural separation and developer guides flow.
11	Learning Concurrency & Cryptography	Study	3.0h	Collaborate	AI acts as an on-demand tutor providing customized analogies and interactive code examples that speed learning.
12	Generating Jest/Supertest Unit Tests	Work	4.0h	Fully Automate	Given a clean controller contract, automated AI test generation reliably produces unit test coverage and edge cases.
13	Project Report & PDF Document Styling	Academic	1.5h	Fully Automate	Document formatting, table rendering, and ReportLab script generation are deterministic tasks that execute reliably.
14	Git Commit Styling & CHANGELOG Generation	Work	1.0h	Fully Automate	Parsing syntax diffs into standardized commit summaries is a structured pattern-matching task suited for automation.
15	Syntax Formatting, Prettier & Linter Fixes	All	1.0h	Fully Automate	Code linting and mechanical formatting rules are purely algorithmic transformations requiring no human cognitive load.

2. Free Toolkit Setup & Anthropic Academy Enrollment Evidence

Tool / Platform	Account Status & Tier	Primary Purpose & Workflow Role
Anthropic Claude	Verified Pro/Free Tier (Claude 3.5)	Primary coding collaborator, architectural reasoning, and Claude Project knowledge base.
OpenAI ChatGPT	Verified Account (GPT-4o)	Cross-verification of SQL/regex patterns and secondary code review perspective.
Anthropic Academy	ENROLLED & MODULE 1 COMPLETED	Course: <i>AI Fluency: Framework & Foundations</i> (FL-01). Applied delegation matrix & guardrails.

3. Configured Claude Project Screenshot & Custom Instructions

Project Title: Full-Stack & Backend AI Intern (small-backend) | **Workspace:** small-backend/workflow
Knowledge Base Files Attached: README.md (Architecture), public.routes.js (Routing), generate_pdf.py (ReportLab)
Custom Instructions Panel (Verified Active):

- **WHO YOU ARE:** Senior Full-Stack & Backend AI Intern assisting Gauta with Node.js/Express REST APIs and Python scripting.
- **tone preferences:** Direct, pragmatic, engineering-first. No introductory filler. Include standard HTTP status codes.
- **CURRENT GOALS:** 1) Build modular REST endpoints in Script/src/. 2) Generate Jest unit tests (>=85% cov). 3) Support PDF reporting.
- **TECHNICAL RULES:** Strict Zod/Joi validation, async/await try/catch boundaries, zero security flaws, preserve existing docstrings.

4. Three Target Audit Tasks for FL-02 through FL-04 (with Success Definitions)

Module	Target Audit Task & Quadrant	Measurable Quantitative Metrics	Qualitative Success Criteria
FL-02 Structured Prompts	Boilerplate Express.js CRUD Routes & Controllers <i>[Delegate to AI with review]</i>	• 70% time reduction: 45 min → <15 min per CRUD endpoint. • 100% first-pass pass rate for ESLint & Zod syntax.	Clean route/controller separation, standard HTTP status codes (200, 201, 400, 404, 500), concise JSDoc headers.
FL-03 Multi-Turn Collab	Debugging Async Race Conditions & DB Pooling <i>[Collaborate with AI]</i>	• 60% triage time reduction: <30 min resolution vs. 2+ hrs. • 0 memory leaks under 500 req/sec load test.	Clear root-cause explanation in PR commit messages, zero happy-path regressions, targeted logging.
FL-04 Automated Pipelines	Automated Jest/Supertest Unit Test Suite Gen <i>[Fully Automate]</i>	• ≥85% branch & function coverage on target controllers. • 100% unattended CI/CD pass rate without flaky mocks.	Explicit assertion of edge cases and auth failure traps (not just happy paths), clean afterEach cleanup.